

TOULON HYERES AP, France

WMO: 076670

Lat: 43.0944N

Lon: 6.1464E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	1.8	-10.0	1.6	4.7	-7.7	2.0	6.0	13.1	11.8	12.1	12.2	2.9	330	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.0	21.0	29.8	21.2	28.8	21.1	24.3	28.2	23.5	27.5	22.7	26.9	5.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	27.3	22.1	16.8	26.5	21.2	15.9	25.8	73.3	28.4	70.0	27.5	67.1	27.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.4	9.2	-1.7	34.5	1.4	1.4	-2.7	35.5	-3.5	36.4	-4.3	37.1	-5.3	38.1		
			-3.2	25.2	1.4	0.8	-4.3	25.7	-5.1	26.2	-5.9	26.7	-6.9	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	9.1	9.0	11.2	13.4	17.0	20.7	23.4	23.4	20.2	17.0	12.6	10.0
	DBStd	5.75	2.61	2.58	2.45	2.22	2.36	2.55	2.14	2.11	2.32	2.47	2.88	2.73
	HDD10.0	152	49	43	14	2	0	0	0	0	0	9	34	
	HDD18.3	1474	287	261	222	148	56	8	1	1	8	171	259	
	CDD10.0	2207	20	17	51	104	216	321	417	416	307	217	89	33
	CDD18.3	487	0	0	1	0	13	78	159	158	65	13	0	0
	CDH23.3	3918	0	0	1	2	66	559	1441	1430	389	31	0	0
	CDH26.7	918	0	0	0	0	7	101	381	380	48	1	0	0
Wind	WSAvg	3.8	3.9	4.0	4.2	4.1	3.8	3.6	3.8	3.5	3.6	3.6	3.9	3.9
Precipitation	PrecAvg	718	77	54	45	64	43	35	11	17	77	106	112	77
	PrecMax	1156	223	149	150	159	101	139	45	80	196	253	331	214
	PrecMin	387	0	0	0	8	0	1	1	0	4	29	24	4
	PrecStd	190	66	46	38	44	30	36	12	18	52	56	74	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	18.0	21.5	22.9	27.4	31.2	33.1	33.0	29.7	25.6	21.0	18.6
		MCWB	12.2	11.2	13.1	14.5	17.7	20.7	19.8	20.9	19.4	18.1	15.6	12.8
	2%	DB	16.3	16.2	18.7	21.1	25.0	29.0	31.0	31.1	27.9	24.1	19.6	16.8
		MCWB	11.0	10.6	12.1	14.3	16.9	20.5	20.8	21.3	19.6	18.2	15.3	12.5
	5%	DB	15.0	15.1	17.2	19.7	23.5	27.7	29.8	29.8	26.8	22.9	18.7	15.8
		MCWB	11.1	10.7	11.7	13.8	16.7	20.4	21.4	21.4	19.7	18.0	14.9	12.3
	10%	DB	14.0	14.1	16.1	18.4	22.2	26.4	28.8	28.8	25.6	21.9	17.8	14.9
		MCWB	10.6	10.3	11.6	13.3	16.2	19.9	21.2	21.5	19.3	17.5	14.4	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.1	14.5	16.7	20.2	24.1	25.2	25.7	23.1	21.0	17.7	14.9
		MCDB	16.0	15.3	18.3	20.0	24.0	27.8	29.0	29.2	26.3	23.3	19.6	16.3
	2%	WB	12.8	12.4	13.7	15.7	18.9	22.8	24.2	24.6	22.1	20.0	16.7	14.0
		MCDB	14.9	14.6	16.8	19.1	22.9	26.9	28.2	28.4	25.8	22.5	18.5	15.7
	5%	WB	12.0	11.7	13.0	14.9	18.1	21.8	23.4	23.7	21.3	19.2	15.9	13.2
		MCDB	14.2	14.1	15.9	18.2	21.8	25.9	27.6	27.7	25.1	21.8	17.9	15.1
	10%	WB	11.1	10.9	12.4	14.2	17.3	20.9	22.7	22.9	20.4	18.3	15.0	12.2
		MCDB	13.5	13.6	15.3	17.3	20.9	25.1	27.0	27.0	24.2	21.1	17.3	14.4
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	9.1	8.9	9.1	9.8	10.0	10.3	9.6	8.8	8.4	8.2
		MCDBR	9.5	10.1	10.5	10.7	10.9	11.0	11.2	11.5	10.8	9.7	9.0	9.1
		MCWBR	6.7	7.2	6.8	6.4	6.1	5.9	6.0	6.5	6.4	6.0	6.2	6.4
	5% WB	MCDBR	7.7	9.0	9.6	9.8	9.9	10.0	10.1	10.4	9.6	8.2	7.4	7.6
		MCWBR	6.3	7.1	6.8	6.6	6.4	6.2	6.5	7.0	6.5	5.8	5.8	5.9
Clear-Sky Solar Irradiance	taub	0.331	0.349	0.385	0.407	0.417	0.416	0.410	0.406	0.398	0.392	0.357	0.327	
	taud	2.476	2.424	2.340	2.296	2.313	2.351	2.371	2.389	2.397	2.399	2.443	2.490	
	Ebn at Noon	802	845	853	859	859	858	859	851	830	782	758	770	
	Edh at Noon	74	91	111	125	127	122	119	113	104	93	76	67	
All-Sky Solar Radiation	RadAvg	1.85	2.75	4.14	5.36	6.43	7.42	7.41	6.45	4.87	3.14	2.02	1.60	
	RadStd	0.18	0.28	0.34	0.35	0.47	0.29	0.29	0.22	0.25	0.25	0.21	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.19	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+44

Nomenclature: See separate page